MATH 1075 - Section 9.1 Worksheet

10/22/2026

1 Solving Absolute Value Equations: $|f(x)| = a$

Solve the equation.

1. $|x| - 14 = -10$

2. $|5x| = 0$

3. $|4x + 1| = 6$

4. $\left|\frac{x}{2} + \frac{3}{2}\right| - 2 = 7$

5. $-3|x - 5| = 7$

6. $5|1 - 2x| - 7 = 3$

2 Solving Absolute Value Equations: $|f(x)| = |g(x)|$

Solve the equation.

1. $|3x + 5| = |-5|$

2. $|3x + 1| = |2x - 7|$

3. $|2x - 6| = |2x + 5|$

4. $\left|\frac{4x-1}{6}\right| = \left|\frac{2x}{3} + \frac{1}{4}\right|$

5. $|4x - 3| = -|2x - 1|$

6. $|x + 2| + |x + 4| = 0$